

3-D parametric shape pass (v0.84.0)

Measured the classifier against the live index, then targeted the real offenders. No guessing — every change is data-driven.

Recognizable shape (vs. box fallback), 24,312 figure-bearing parts



= 3,736 more parts now render as a recognizable shape instead of a blank box.

Added 9 geometry families (22 total; all verified to build valid meshes)

plate flat slab + bolt-holes markers · decals · labels · armor	cover open panel / pan covers · doors · panels · guards	pad cushion slab insulation · pads · straps
link bar + two end eyes connecting links	lever arm + pivot + grip levers · handles · cranks	rivet shank + button head solid / blind rivets
switch body + toggle + pins switches · relays	cylinder body + rims + port motors · pumps · actuators · valves	canister domed cylinder air cleaners · filters · tanks

Also fixed

A real bug: CLAMP was read as a shaft because the pattern 'LAMP' had no word boundary (matched cLAMP). Bounded LAMP/BULB/CAP/STUD/COIL. Residual box is large assemblies (engine, van, frame) that carry the real manual figure anyway. partgeo.js ?v= bumped so browsers reload.